

Vipul Harsh

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Interests	Current interests: Failure diagnosis in networked systems, reliability of AI agents, network topology Broad interests: Networked and Distributed Systems, ML for Systems, Algorithms	
Experience	Postdoctoral researcher, Conviva, Foster City, California	July 2024 - Ongoing
	Visiting researcher, CMU Advisors: Vyas Sekar , Hui Zhang	
	Research intern, VMware Research, Palo Alto, California (remote) Advisors: Sujata Banerjee , Radhika N. Mysore	June 2020 - Sept. 2020
Education	Software engineering intern, Google, Sunnyvale, California Advisors: Gautam Kumar , Nandita Dukkupati	May 2018 - August 2018
	University of Illinois at Urbana-Champaign Ph.D., Computer Science Advisor: P. Brighten Godfrey	2017 - 2024
	University of Illinois at Urbana-Champaign M.S., Computer Science Advisor: Laxmikant Kale	2015 - 2017
	Indian Institute of Technology, Bombay B.Tech. (Honors), Computer Science and Engineering	2011 - 2015
Publications	MoCE: A Mixture of Context Aware Experts Framework for Troubleshooting Internet-scale Services , NSDI 2026 <i>Vipul Harsh, Sayan Sinha, Henry Milner, B. Aditya Prakash, Vyas Sekar, Hui Zhang</i>	
	Starfish: A Topology-Routing Co-Design for Small-Scale Data Centers , NSDI 2026 Anchengcheng Zhou, Vipul Harsh , Sangeetha A. Jyothi, Maria Apostolaki, P. Brighten Godfrey	
	Automatically Surfacing Opportunities for Improvements In Internet-Scale Applications , HotNets 2025 <i>Vipul Harsh, Sayan Sinha, Henry Milner, Haijie Wu, B. Aditya Prakash, Vyas Sekar, Hui Zhang</i>	
	TraceWeaver: Distributed Request Tracing for Microservices Without Application Modification , SIGCOMM 2024 <i>Sachin Ashok, Vipul Harsh, P. Brighten Godfrey, Radhika Mittal, Srinivasan Parthasarathy, Larisa Shwartz</i>	
	Murphy: Performance Diagnosis of Distributed Cloud Applications , SIGCOMM 2023 <i>Vipul Harsh, Wenxuan Zhou, Sachin Ashok, Radhika N. Mysore, P. Brighten Godfrey, Sujata Banerjee</i> VMware's blog post about product adoption: https://shorturl.at/efvT2	
	Flock: Accurate Datacenter Fault Localization at Scale , CoNEXT 2023 <i>Vipul Harsh, Tong Meng, Kapil Agrawal, P. Brighten Godfrey</i>	
	Optimal Round and Sample-Size Complexity for Partitioning in Parallel Sorting , SPAA 2023 <i>Wentao Yang*, Vipul Harsh*, Edgar Solomonik (*: equal contribution)</i>	
	Spineless Datacenters , HotNets 2020 <i>Vipul Harsh, Sangeetha A. Jyothi, P. Brighten Godfrey</i>	
	Histogram Sort with Sampling , SPAA 2019 <i>Vipul Harsh, Laxmikant Kale, Edgar Solomonik</i>	
	There's Waldo: Localizing failures among symmetric components Vipul Harsh, Rahul Bothra, P. Brighten Godfrey	
Manuscripts in preparation		

Patents

- **On-demand Network Incident Graph Generation** (US Patent App. 18/094,378)
Vipul Harsh, Wenxuan Zhou, Radhika Niranjana Mysore, Philip Brighten Godfrey, Sujata Banerjee
- **Network Incident Root-Cause Analysis** (US Patent App. 18/094,379)
Vipul Harsh, Wenxuan Zhou, Radhika Niranjana Mysore, Philip Brighten Godfrey, Sujata Banerjee
- **Providing Explanation of Network Incident Root Causes** (US Patent App. 18/094,380)
Vipul Harsh, Wenxuan Zhou, Radhika Niranjana Mysore, Philip Brighten Godfrey, Sujata Banerjee

Awards

- Selected for the [NSF-NetS early career workshop](#), January 2025 at Alexandria, VA, USA
- Represented IIT Bombay at the ACM ICPC World Finals 2015. Highest ranked team from India
- Ranked 49 in IIT-JEE 2011, amongst 500,000 candidates
- Rank 1 in International Mathematics Olympiad, 2009 conducted by Science Olympiad Foundation

Internships

- **Research Internship, UC Berkeley** June 2019 - August 2019
Distributed garbage collection for actor-based systems
- **Research Internship, Georgia Tech** May 2014 - July 2014
Large scale simulations for polydisperse hydrodynamic particle systems
- **Research Internship, LaBRI, France** May 2013 - July 2013
Algorithms for computing coverability sets of Petri Nets

Teaching

- Teaching Assistant, Cloud Networking, UIUC (Spring 2024)
- Teaching Assistant, Probability and Statistics for Computer Science, UIUC (Spring 2022)
- Teaching Assistant, Discrete Mathematics, IIT Bombay
- Teaching Assistant, GPU Programming and Applications Workshop (GPA), IIT Bombay

Talks

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| • HotNets 2025 , College Park, Maryland
Automatically Surfacing Opportunities for Improvements In Internet-Scale Applications | November 2025 |
| • CMU , Pittsburgh, Pennsylvania
Abstractions for high-coverage, extensible and scalable root cause analysis (RCA)
Hosted by Prof. Vyas Sekar | November 2025 |
| • MIT , Cambridge, Massachusetts
Failure Diagnosis in Networked systems
Hosted by Prof. Christina Delimitrou | July 2024 |
| • Conviva , Foster city, California
Failure Diagnosis in Networked systems
Hosted by Prof. Vyas Sekar | March 2024 |
| • CoNEXT 2023 , Paris
Flock: Accurate Network Fault Localization at Scale | December 2023 |
| • SIGCOMM 2023 , New York City
Murphy: Performance Diagnosis of Distributed Cloud Applications | September 2023 |
| • SPAA 2023 , Orlando, Florida
Optimal Round and Sample-Size Complexity for Partitioning in Parallel Sorting | June 2023 |
| • EnvoyCon at KubeCon , Detroit, Michigan
Distributed Tracing without the pain | October 2022 |
| • VMware Research , Palo Alto, California
Murphy: Performance diagnosis of Distributed Cloud Applications | August 2022 |
| • VMware RADIO , San Francisco, California
Murphy: Performance diagnosis of Distributed Cloud Applications | May 2022 |
| • HotNets 2020 , Chicago, Illinois
Spineless Data Centers | November 2020 |
| • VMware , Palo Alto, California
Fast and accurate datacenter fault localization | February 2020 |
| • Google , Sunnyvale, California
Fast and accurate datacenter fault localization | August 2019 |

- **SPAA 2019**, Phoenix, Arizona
Histogram sort with sampling June 2019
- **Charm++ Workshop**, Urbana, Illinois
Histogram sort with sampling April 2017

References
(on request)

- **Vyas Sekar**, CMU, Conviva
- **Hui Zhang**, CMU, Conviva
- **Brighten Godfrey**, UIUC
- **Sujata Banerjee**, Microsoft Research
- **Edgar Solomonik**, UIUC

Service

- Program Committee, Conext 2025